Mississippi Valley State University

CS 341 — Discrete Structures

Practice Set 48

Divide-and-Conquer Recurrences

Covers *MCS* Sections 21.4 and 21.5

Status of this set

Ungraded practice. Solutions released with it. **Always state which case you are in** — that is the graded step.

Part I: How to Practice This Material

The Master Theorem as a competition

For $T(n) = a\,T(n/b) + g(n)$, compute $p = \log_b a$ and compare $g(n)$ with n^p.

Which is bigger	Case	Answer
n^p wins	1	$\Theta(n^p)$
they tie	2	$\Theta(n^p \log n)$
$g(n)$ wins	3	$\Theta(g(n))$

Compute p *first*. Everything follows from comparing it with g.

Three mistakes to expect

1. **Computing $\log_a b$ instead of $\log_b a$.** It is (number of subproblems) in the exponent position.
2. **Forgetting the $\log n$ in Case 2.**
3. **Using the Master Theorem on $f(n-1)$ recurrences.** Those need Wednesday's characteristic equation.

Part II: Practice Problems

Section A — Applying the theorem

1. For each, give $p = \log_b a$, the case, and the solution.

(a) $T(n) = 9T(n/3) + n$

(b) $T(n) = 2T(n/2) + 1$

(c) $T(n) = 8T(n/2) + n^3$

(d) $T(n) = 3T(n/4) + n$

2. Identify the two algorithms and confirm their complexities with the Master Theorem.

(a) $T(n) = T(n/2) + 1$

(b) $T(n) = 2T(n/2) + n$

Section B — Which method?

3. For each, say whether the characteristic-equation method or the Master Theorem applies, and why. Do not solve.

(a) $f(n) = 4f(n-1) - 3f(n-2)$

(b) $T(n) = 5T(n/5) + n$

(c) $f(n) = f(n-1) + 2^n$

(d) $T(n) = T(n-1) + T(n/2)$

Section C — The comparison

4. Towers of Hanoi satisfies $T(n) = 2T(n-1) + 1$ and merge sort satisfies $T(n) = 2T(n/2) + n$. Both split into two subproblems.

(a) Which does more work per recursive call?

(b) Which has the smaller solution, and by how much at $n = 64$?

(c) What does the comparison show?

Section D — Writing

5. In five to eight complete sentences, explain why the Master Theorem compares $g(n)$ with $n^{\log_b a}$ rather than with n or with a. What does $n^{\log_b a}$ represent?